Fine-Grained Bird Species Classification - CUB-200-2011

Two models were developed for 200-class fine-grained classification using the CUB-200-2011 dataset. Only the official training set was used for model development. A stratified 90:10 split was applied to create a validation subset to ensure balanced class representation. Random seeds were fixed to ensure reproducibility.

Data Processing

Bounding box annotations were used to crop images around the bird region with additional padding (15%) to retain contextual features. This was implemented to reduce background noise and improve fine grained feature learning. Images were resized to 260×260 to balance computational cost and detail preservation. EfficientNet-style normalization was applied. Data augmentation included horizontal flipping, rotation, contrast adjustments and MixUp (alpha = 0.1). MixUp was selected after observing reduced overfitting compared to standard augmentation alone.

Model 1 - Transfer Learning (EfficientNetB2)

EfficientNetB2 was chosen due to its compound scaling strategy, which balances depth, width and resolution for efficient feature extraction. Fine-grained bird classification requires capturing subtle visual differences, and EfficientNet's squeeze-and-excitation blocks enhance channel attention for discriminative features. B2 provides a good trade-off between accuracy and computational cost, while pretrained ImageNet weights improve convergence and generalization.

Initially, EfficientNetB0 and EfficientNetB3 were tested out. But they did not give the best results as EfficientNetB2. So, An EfficientNetB2 backbone pretrained on ImageNet was used. Transfer learning was chosen due to strong prior performance on fine-grained tasks. The backbone was initially frozen to train the classifier head, followed by partial fine-tuning of the last 40 layers. A dropout rate of 0.30 was selected after testing 0.2 and 0.5, with 0.30 providing the best validation performance.

The model was trained using AdamW (LR = 1e-3 for head training, 5e-6 cosine decay for fine-tuning). Learning rates were tuned experimentally by testing 1e-2, 1e-3, and 1e-4, where 1e-3 showed fastest stable convergence. Mixed precision training was used for efficiency.

Test Performance (Flip-TTA):
Accuracy: 80.20%
Precision (macro): 0.835
Recall (macro): 0.802

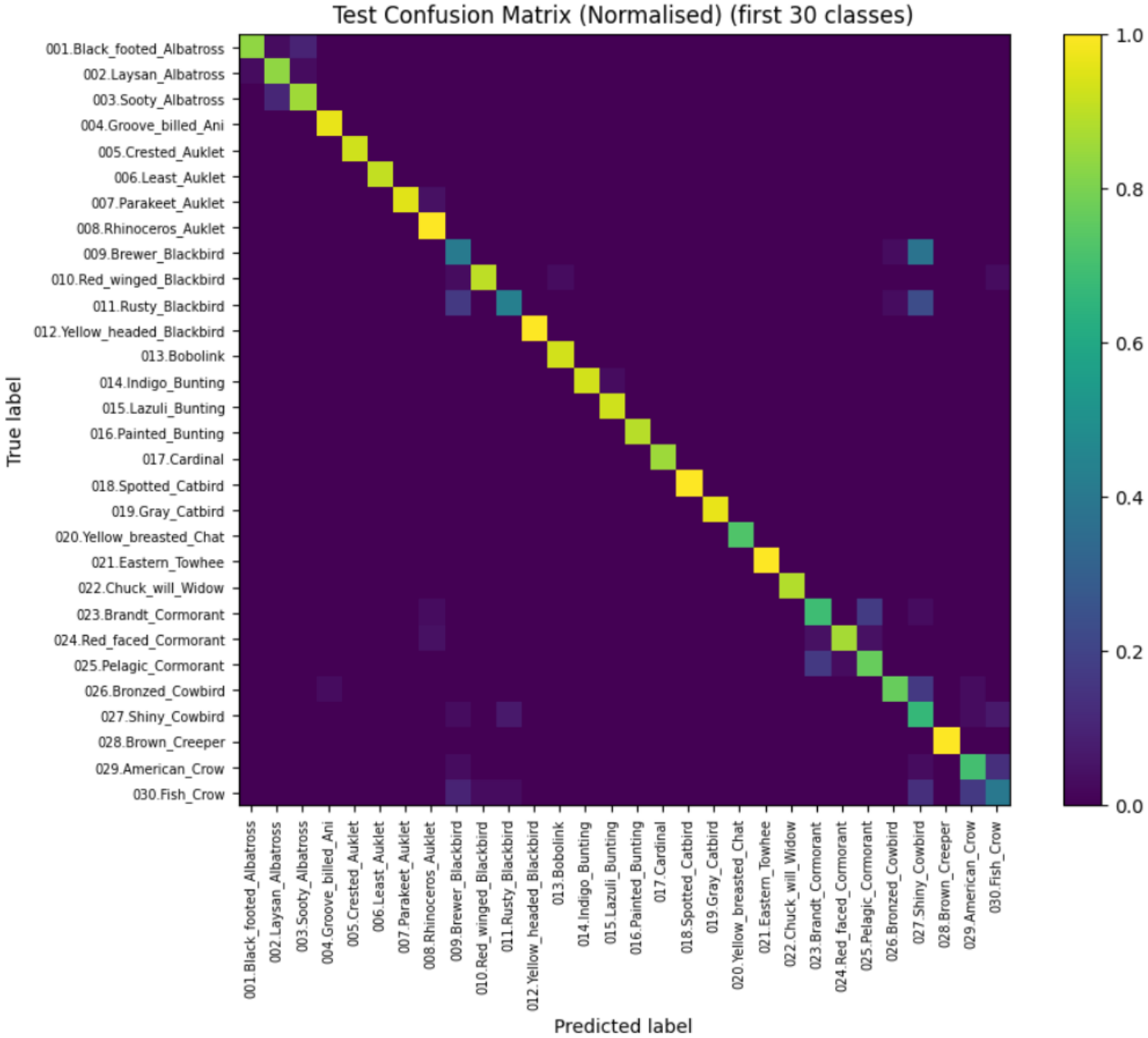
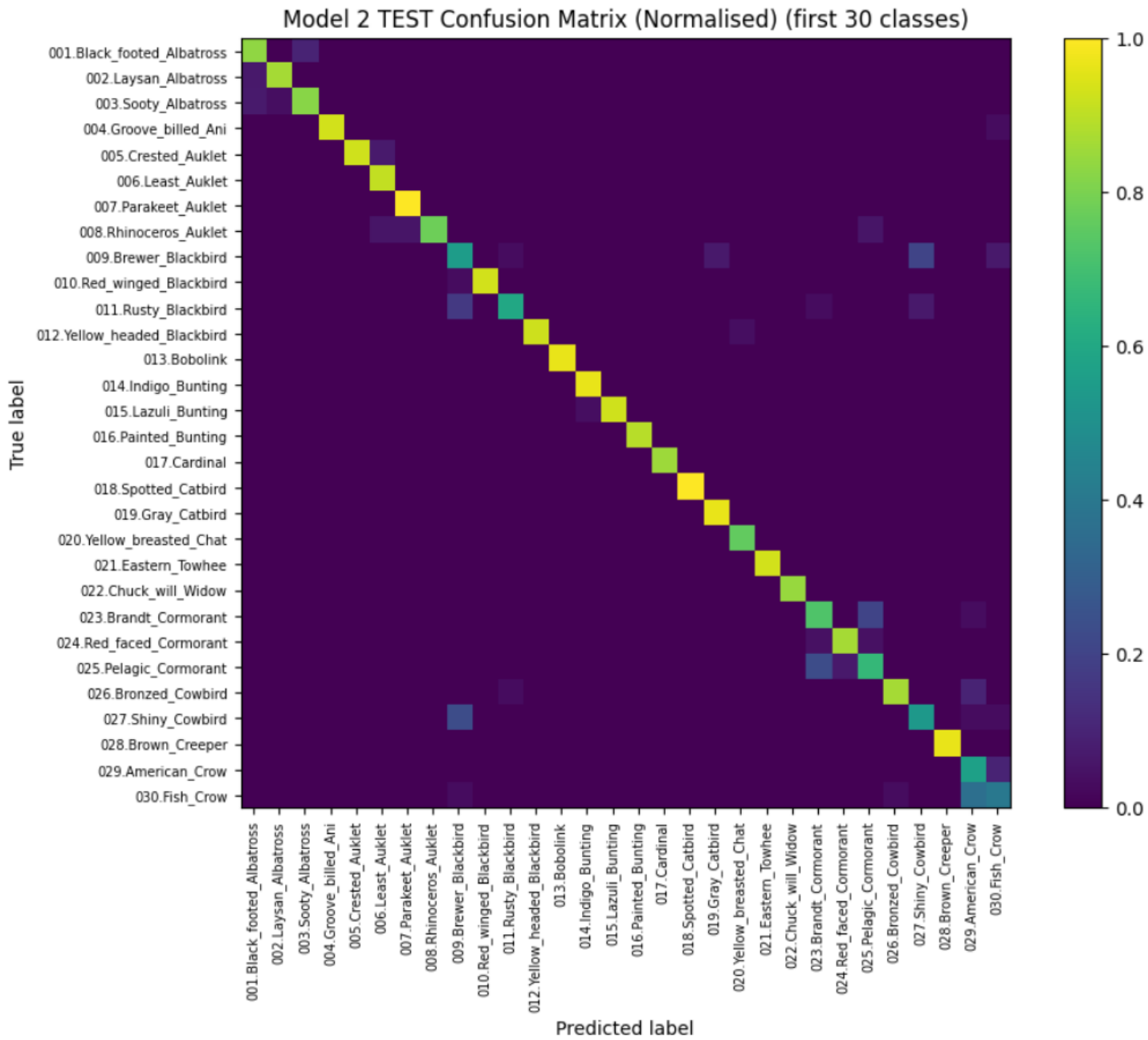
Model 2 - Custom CNN with Attention

A custom convolutional network (~1.6M parameters) with channel attention was designed to focus on discriminative regions. ImageNet initialization was used to stabilize training. Dropout (0.40) was selected after testing 0.3 and 0.5. The initial learning rate of 3e-4 was selected after observing unstable gradients at 1e-3 and slow convergence at 1e-4. ReduceLROnPlateau was applied based on validation loss.

Test Performance (Flip-TTA):
Accuracy: 79.60%
Precision (macro): 0.808
Recall (macro): 0.798

Confusion Matrices

Model 1 - EfficientNetB2



Model 2 - Custom CNN with Attention